

Assignment-4

Parallel & Distributed Computing
(UCS645)

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Performance Analysis

Ring Communication

Processes (p)	Time (T _p) (s)	Speedup (S _p)	Efficiency (E _p)
1	6.2544e-05	1.00	1.00
2	8.6308e-05	0.72	0.36
4	3.45825e-04	0.18	0.045
8	3.62997e-03	0.017	0.002

Table 1: Performance of Ring Communication

Observations:

- Execution time increases with number of processes.
- Speedup is less than 1, indicating performance degradation.
- Communication overhead dominates computation.
- Efficiency drops drastically as number of processes increases.
- Ring communication is communication-heavy rather than computation-heavy.

Parallel Array Sum

Processes (p)	Time (T _p) (s)	Speedup (S _p)	Efficiency (E _p)
1	4.007e-06	1.00	1.00
2	1.2767e-05	0.31	0.155
4	2.01e-04	0.019	0.0047
8	1.74033e-03	0.0023	0.00029

Table 2: Performance of Parallel Array Sum

Observations:

- Performance degrades as number of processes increases.
- Communication overhead exceeds computation time.
- Incorrect result at 8 processes indicates uneven data division.
- Small problem size makes parallel execution inefficient.
- Efficiency is extremely low for higher number of processes.

Max-Min Computation

Observations:

- Execution time increases significantly with more processes.
- Very low speedup due to minimal computation.
- MPI_Reduce overhead dominates performance.

Processes (p)	Time (T _p) (s)	Speedup (S _p)	Efficiency (E _p)
1	2.747e-06	1.00	1.00
2	2.0794e-05	0.13	0.065
4	3.86828e-04	0.0071	0.0018
8	3.217e-04	0.0085	0.0011

Table 3: Performance of Max-Min Program

- Slight variation at higher processes due to system effects.
- Efficiency remains very low across all parallel executions.

Dot Product

Processes (p)	Time (T _p) (s)	Speedup (S _p)	Efficiency (E _p)
1	4.073e-06	1.00	1.00
2	7.641e-06	0.53	0.265
4	5.6382e-05	0.072	0.018
8	1.2802e-04	0.032	0.004

Table 4: Performance of Dot Product

Observations:

- Slight improvement from 1 to 2 processes.
- Performance decreases beyond 2 processes.
- Communication cost outweighs computation.
- Small vector size results in poor scalability.
- Efficiency decreases rapidly with increase in processes.

Overall Analysis

- Parallel execution does not always guarantee better performance.
- For small problem sizes, communication overhead dominates computation.
- Speedup less than 1 indicates inefficient parallelization.
- Efficiency decreases sharply with increasing number of processes.
- Larger problem sizes are required to fully utilize parallel computing.
- MPI is most effective when computation cost is significantly higher than communication cost.